CHAPTER - I

# CRYSTAL STRUCTURE

## 1.1 INTRODUCTION

Matter, consisting of one or more elements or their chemical compounds, exists in nature in the solid, liquid and gaseous states. As the atoms or molecules in solids are attached to one another with strong forces of attraction, the solids maintain a definite volume and shape. The solid state physics is the branch of physics dealing with physical properties of solids, particularly crystals, including the behaviour of electrons in these solids. The solids may be broadly classified as *crystalline* and *non-crystalline* depending upon the arrangement of atoms or molecules.

The crystalline state of solids is characterized by regular or periodic arrangement of atoms or molecules. Most of the solids are crystalline in nature. This is due to the reason that the energy released during the formation of an ordered structure is more than that released during the formation of a disordered structure. Thus crystalline state is a low energy state and is, therefore, preferred by most of the solids. The crystalline solids may be subdivided into *single crystals* and *polycrystalline* solids. In single crystals, the periodicity of atoms extends throughout the material as the case of diamond, quartz, mica, etc. A polycrystalline material is an aggregate of a number of small crystallites with random orientations separated by well-defined boundaries. The small crystallites are known as *grains* and the boundaries as *grain boundaries.* It may be noted that although the periodicity of individual crystallites is interrupted at grain boundaries, yet the polycrystalline form of a material may be more stable compared with its single crystal form. Most of the metals and ceramics exhibit polycrystalline structure.

The non-crystalline or *amorphous* solids are characterized by the completely random arrangement of atoms or molecules. The periodicity, if at all present, extends up to a distance of a few atomic diameters only. In other words, these solids exhibit *short range order.* Such type of materials are formed when the atoms do not get sufficient time to undergo a periodic arrangement. Glass is an example of amorphous materials. Most of the plastics and rubbers are also amorphous.

The science which deals with the study of geometrical forms and

physical properties of crystalline solids is called *crystallography*. The study of crystallography is necessary to understand the strong correlation between the structure of a material and its physical properties. The present chapter deals with some of the basic concepts of crystallography which are fundamental to the study of solid state physics.

## 1.2 CRYSTAL LATTICE AND TRANSLATION VECTORS

Before describing the arrangement of atoms in a crystal, it is always convenient to describe the arrangement of imaginary points in space which has a definite relationship with the atoms of the crystal. This set of imaginary points forms a framework on which the actual crystal structure is based. Such an arrangement of infinite number of imaginary points in three-dimensional space with each point having identical surroundings is known as *point lattice* or *space lattice*.

The term 'identical surroundings' means that the lattice has the same appearance when viewed from a point **r** in the lattice as it has when viewed from any other point **r'** with respect to some arbitrary origin. This is possible only if the lattice contains a small group of points, called *pattern unit*, which repeats itself in all directions by means of a *translation operation* **T** given by

$$\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c} \tag{1.1}$$

where $n_1$, $n_2$ and $n_3$ are arbitrary integers and the vectors **a**, **b** and **c** are called the *fundamental translation vectors*. Thus, we have

$$\mathbf{r'} = \mathbf{r} + \mathbf{T} = \mathbf{r} + n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c} \tag{1.2}$$

In a perfect lattice, Eq. (1.2) holds good, i.e., point **r'** can be obtained from **r** by the application of the operation (1.1). However, in an imperfect lattice, it is not possible to find **a**, **b** and **c** such that an arbitrary choice of $n_1$, $n_2$ and $n_3$ makes **r'** identical to **r**. The translation vectors **a**, **b** and **c** are also called the *crystal axes* or *basis vectors* and shall be described later.

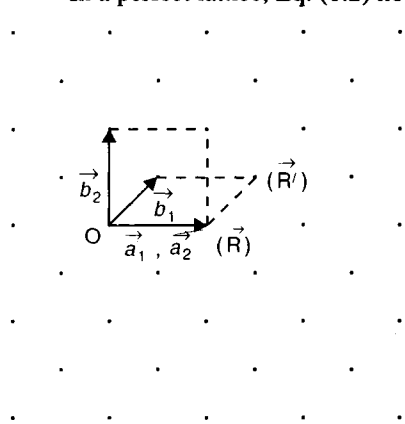



Fig. 1.1. *Primitive* $(\mathbf{a}_1, \mathbf{b}_1)$ *and non-primitive* $(\mathbf{a}_2, \mathbf{b}_2)$ *translation vectors in a two-dimensional lattice.*

Consider, for simplicity, a part of a two-dimensional lattice as shown in Fig 1.1. The translation vectors **a** and **b** can be chosen in a number of ways. Two such possibilities are shown in this figure

where two sets $\mathbf{a}_1$, $\mathbf{b}_1$ and $\mathbf{a}_2$, $\mathbf{b}_2$ of translation vectors are drawn. Considering first the translation vectors $\mathbf{a}_1$ and $\mathbf{b}_1$, the point **R'** can be obtained from **R** using the translation operation given by

$$\mathbf{T} = 0\mathbf{a}_1 + 1\mathbf{b}_1$$

which contains integral coefficients. Thus **R'** is related to **R** by the equation

$$\mathbf{R'} = \mathbf{R} + \mathbf{T} = \mathbf{R} + 0\mathbf{a}_1 + 1\mathbf{b}_1$$

Such translation vectors which produce a translation operation containing integral coefficients are called *primitive translation vectors.* Referring to the second set of translation vectors $\mathbf{a}_2$ and $\mathbf{b}_2$, the point **R'** can be obtained from **R** by using the equation

$$\mathbf{R'} = \mathbf{R} + \frac{1}{2}\mathbf{a}_2 + \frac{1}{2}\mathbf{b}_2$$

which contains non-integral coefficients of $\mathbf{a}_2$ and $\mathbf{b}_2$. Such translation vectors for which the translation operation contains non-integral coefficients are called *non-primitive translation vectors.* Either type of translation vectors may be used to describe the structure of a crystal inspite of the fact that the non-primitive translation vectors involving non-integral coefficients are not in accordance with the periodicity of the crystal. Usually, a set of orthogonal and the shortest possible translation vectors is preferred for describing a lattice.

## 1.3 UNIT CELL

The parallelograms formed by the translation vectors (Fig. 1.1) may be regarded as building blocks for constructing the complete lattice and are known as *unit cells* of the lattice. For a three-dimensional lattice, the unit cells are of the form of a parallelopiped. An application of the translation operation (1.1) for some values of $n_1$, $n_2$ and $n_3$ takes the unit cell to another region which is exactly similar to the initial region. On repeatedly applying the same operation with all possible values of $n_1$, $n_2$ and $n_3$, one can reproduce the complete lattice. Thus a unit cell may be defined as the smallest unit of the lattice which, on continuous repitition, generates the complete lattice. Both primitive and non-primitive translation vectors may be used to construct a unit cell. Accordingly, a unit cell is named as a *primitive unit cell* or a *non-primitive unit cell.* In Fig. 1.2, the parallelogram ABCD represents a two-dimensional primitive cell, whereas the parallelograms EFGH and KLMN represent non-primitive cells. Primitive unit cell is the smallest volume cell. All the lattice points belonging to a primitive cell lie at its corners. Therefore, the effective number of lattice points in a primitive unit cell is one. A non-primitive cell may have the lattice points at the corners as well as at other locations both

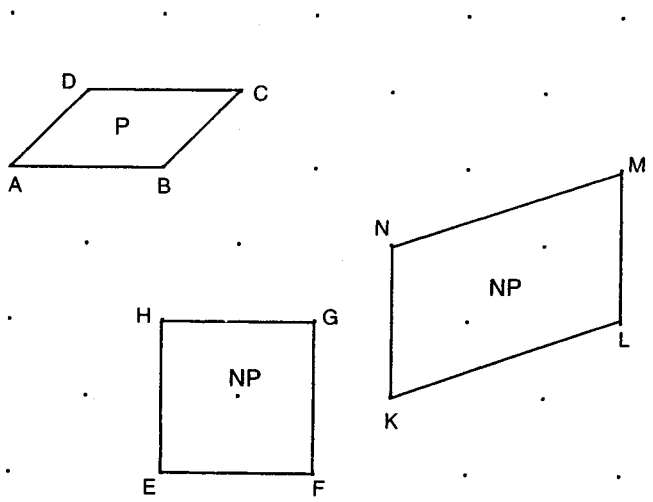



Fig. 1.2. *Primitive (P) and non-primitive (NP) unit cells of a two-dimensional lattice.*

inside and on the surface of the cell and, therefore, the effective number of lattice points in a non-primitive cell is greater than one. A primitive cell can also be constructed using the following procedure :

(i) Connect a given lattice point to all the nearby lattice points.

(ii) Draw normals at the mid-points of lines connecting the lattice points.

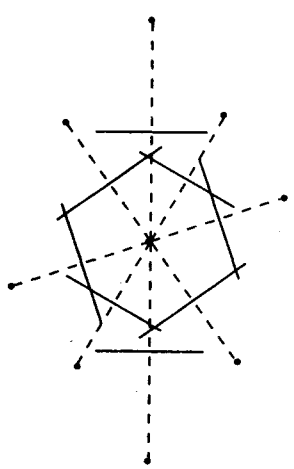


Fig. 1.3. *Construction of Wigner-Seitz primitive cell.*

The smallest volume enclosed by the normals is the required primitive cell. Such a cell is called *Wigner-Seitz cell* and is shown in Fig. 1.3. The volume of a primitive cell having **a, b** and **c** as the fundamental translation vectors or crystallographic axes is given by

$$V = |\mathbf{a}.\mathbf{b}\times\mathbf{c}|$$

Since there exists a number of ways of choosing a unit cell, the choice of a *conventional unit cell* is a matter of convenience. Ideally, the primitive cell having the smallest volume should be chosen as the conventional unit cell.